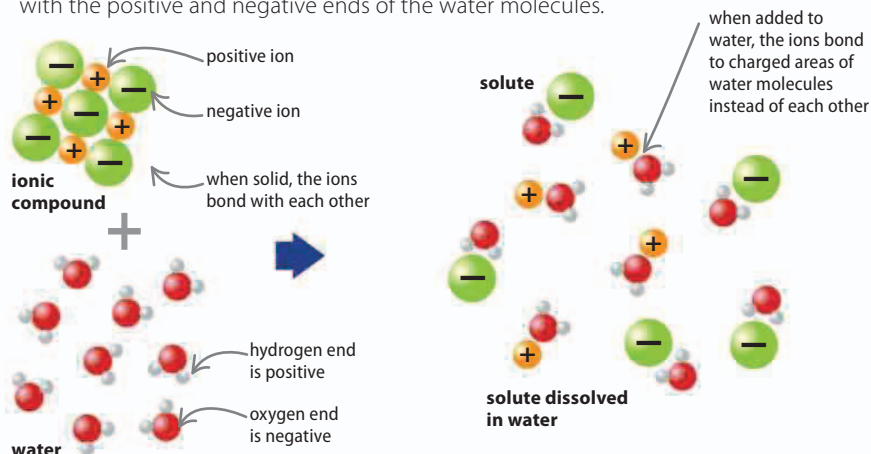


Universal solvent

Water is sometimes known as the universal solvent because so many substances dissolve in it. The property is another result of water molecules' polarity. Ionic substances are made up of charged particles bonded together. When they are added to water, the ions split from their opposite partners and form bonds with the positive and negative ends of the water molecules.



REAL WORLD

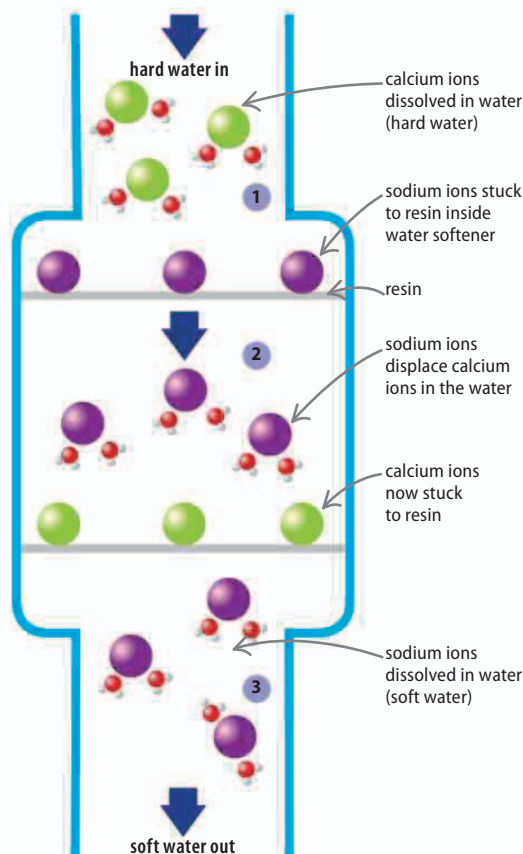
The Dead Sea

When things dissolve in water they make it more dense. Seawater is more dense than freshwater because it has salt dissolved in it. The water in the Dead Sea is so salty that it is much denser than the human body. That is why bathers can float so easily.



Water hardness

Hardness is the term used to describe how many minerals are dissolved in water. Temporary hardness is largely due to dissolved calcium hydrogen carbonate. When the water is heated it decomposes into carbon dioxide, water, and solid calcium carbonate. This solid is called limescale and builds up on heating equipment like kettles. Other calcium (and magnesium) compounds form permanent hardness. They can affect the taste of drinking water and the action of soaps. A water softener replaces minerals that cause hardness with sodium ions, which do not cause as many problems.



1. Hard water in

Dissolved calcium ions make the water hard. This hard water is fed into the softener before it reaches the tap.

2. Inside the softener

The softener contains a porous resin filled with sodium ions. As the hard water flows through, the sodium ions displace (swap places with) the less reactive calcium ions in the water.

3. Soft water out

The water flowing out of the softener contains sodium ions, while the calcium ions are left behind in the resin. The resin needs to be replaced regularly or washed with a sodium solution to restock the sodium ions.